



Basic Tutorial

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Home Module

	S1	S2	S3	S4	S5	S6
P1	5	16	5	0	0	3
P2	21	45	58	14	10	15

Peak Count Data

C

Data Filtering and Grouping Module

Group1

S1

S2

S3

Group2

S4

S5

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D

Differential Peak Analysis Module

	log2FC	pvalue	padj
P1	-2.93	0.04	0.39
P2	-1.35	0.02	0.26

Differential Peak Table

B

Peaks Visualization Module

Coverage Plot

Profile Plot

E

Differential Peak Annotation Module

	anno	TSS	geneID
P1	Exon	779943	114798
P2	Intron	283818	2567

Peak Annotation Table

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Group1 vs Group2

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Import Data

Data Source:

Example dataset

Example dataset

Upload data

dexamethasone treatment, obtained from ENCODE (ENCSR783SNV and ENCSR543ZVZ). Each condition contains three biological replicates, and the peaks have been pre-processed into a merged count matrix ready for analysis. You can load this dataset to explore the features and workflow of ChromTag without uploading your own files.

START

Instruction

The Home Module serves as the entry point of ChromTag, providing an integrated interface for initializing data input and establishing the foundation for subsequent analyses. Users may choose to begin with the system-provided example dataset or upload custom peak count matrices that have been pre-merged across samples. This module also manages species selection and displays an immediate preview of the imported dataset, enabling users to verify data structure and completeness before progressing to downstream analytical modules. Together, these components ensure that the analysis pipeline begins with properly formatted and biologically consistent input data, thereby supporting accurate and reliable interpretation in later stages.

Users can choose between “**Example Dataset**” and “**Upload Data**”.

If the **Example Dataset** is selected, the application automatically loads a **pre-processed merged peak count matrix** containing H3K27ac ChIP-seq profiles from A549 cells collected at 0 h and 4 h after dexamethasone treatment. The dataset was obtained from ENCODE, including ENCSR783SNV and ENCSR543ZVZ, with **three biological replicates** for each condition.

Home Module

 Import Data

Data Source:

Upload data

Select Species:

Human

Upload CSV File

Browse...

No file selected

If you choose to upload custom data, you must also select the appropriate species.

Chromosome	Start	End	A549_0h_rep1	A549_0h_rep2	A549_0h_rep3	A549_4h_rep1	A549_4h_rep2	A549_4h_rep3
chr1	629727	630151	31	12	5	19	18	23
chr1	633792	634272	44	24	10	17	48	25
chr1	778021	778694	46	54	59	69	26	64
chr1	778877	779438	69	53	57	66	37	70
chr1	779666	779906	20	18	22	23	12	29
chr1	826707	827761	97	90	94	88	47	100
chr1	870010	870297	9	17	4	12	6	12
chr1	904028	905405	205	231	182	178	107	185
chr1	906957	907195	28	31	38	18	13	24

Uploaded data should be **pre-merged across samples** and must follow a specific format. The file should be a **CSV** with a header row.

The first column must contain the chromosome name (e.g., “chr1”), the second and third columns should specify the **start** and **end** positions, and all subsequent columns should represent the **peak count values** for each sample.

Each experimental group must include **at least two biological replicates** to ensure proper statistical analysis.

Home Module

Import Data

Data Source:

Example dataset

Example dataset

Upload data

dexamethasone treatment, obtained from ENCODE (ENCSR783SNV and ENCSR543ZVZ). Each condition contains three biological replicates , and the peaks have been pre-processed into a merged count matrix ready for analysis. You can load this dataset to explore the features and workflow of ChromTag without uploading your own files.

▶ START

After choosing the data source, users can click the **“Start”** button to load the dataset and proceed to the next module.

Data Preview

Show 10 entries

Search:

	Chromosome	Start	End	A549_0h_rep1	A549_0h_rep2	A549_0h_rep3	A549_4h_rep1	A549_4h_rep2	A549_4h_rep3
1	chr1	629727	630151	31	12	5	19	18	23
2	chr1	633792	634272	44	24	10	17	48	25
3	chr1	778021	778694	46	54	59	69	26	64
4	chr1	778877	779438	69	53	57	66	37	70
5	chr1	779666	779906	20	18	22	23	12	29
6	chr1	826707	827761	97	90	94	88	47	100
7	chr1	870010	870297	9	17	4	12	6	12
8	chr1	904028	905405	205	231	182	178	107	185
9	chr1	906957	907195	28	31	38	18	13	24
10	chr1	910946	911195	18	17	15	13	6	21

Showing 1 to 10 of 76,176 entries

Previous

1

2

3

4

5

...

7,618

Next

Once the data is successfully loaded, the system will display the **peak count matrix** in a preview table for inspection.

Peaks Visualization Module

This step is optional and can be skipped.

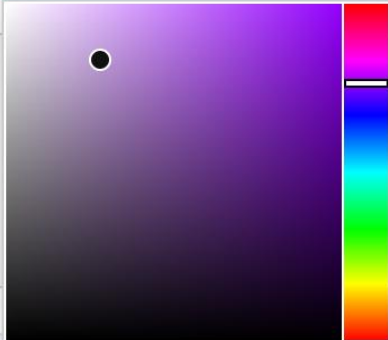
 Assign Sample Colors

Sample:

H3K4me3_rep2 ▼

Choose a Color:

#BE9CD6



✓ Submit

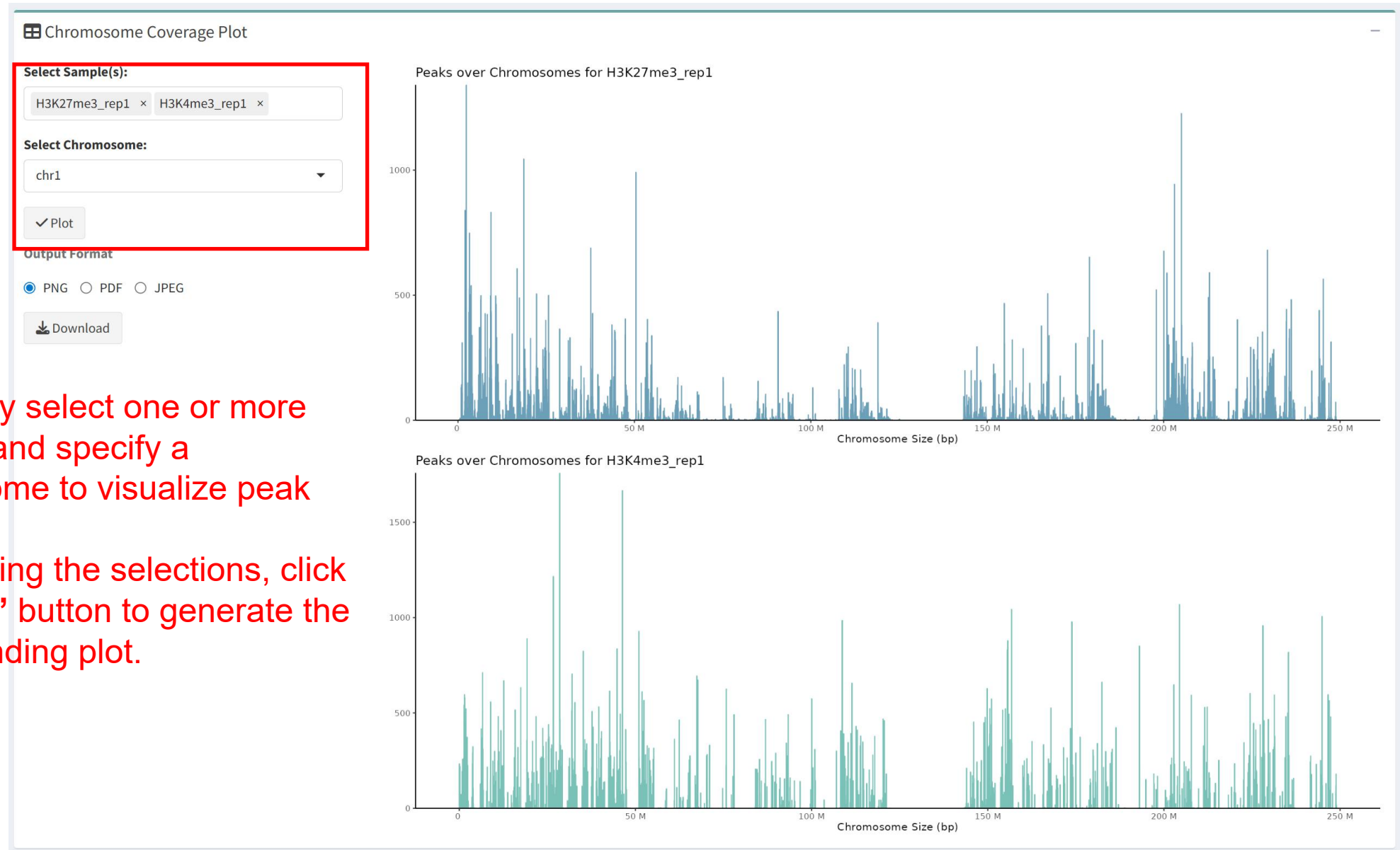
Sample	Color
H3K27me3_rep1	#6C9FB8
H3K27me3_rep2	#A587C9
H3K4me3_rep1	#7AC2B8
H3K4me3_rep2	#C5D192

Users can select a sample and assign a preferred color using the color picker. Once submitted, the chosen color will be applied consistently across the **Chromosome Coverage Plot** and the **TSS Profile Plot**, enhancing visual distinction among samples.

Peaks Visualization Module

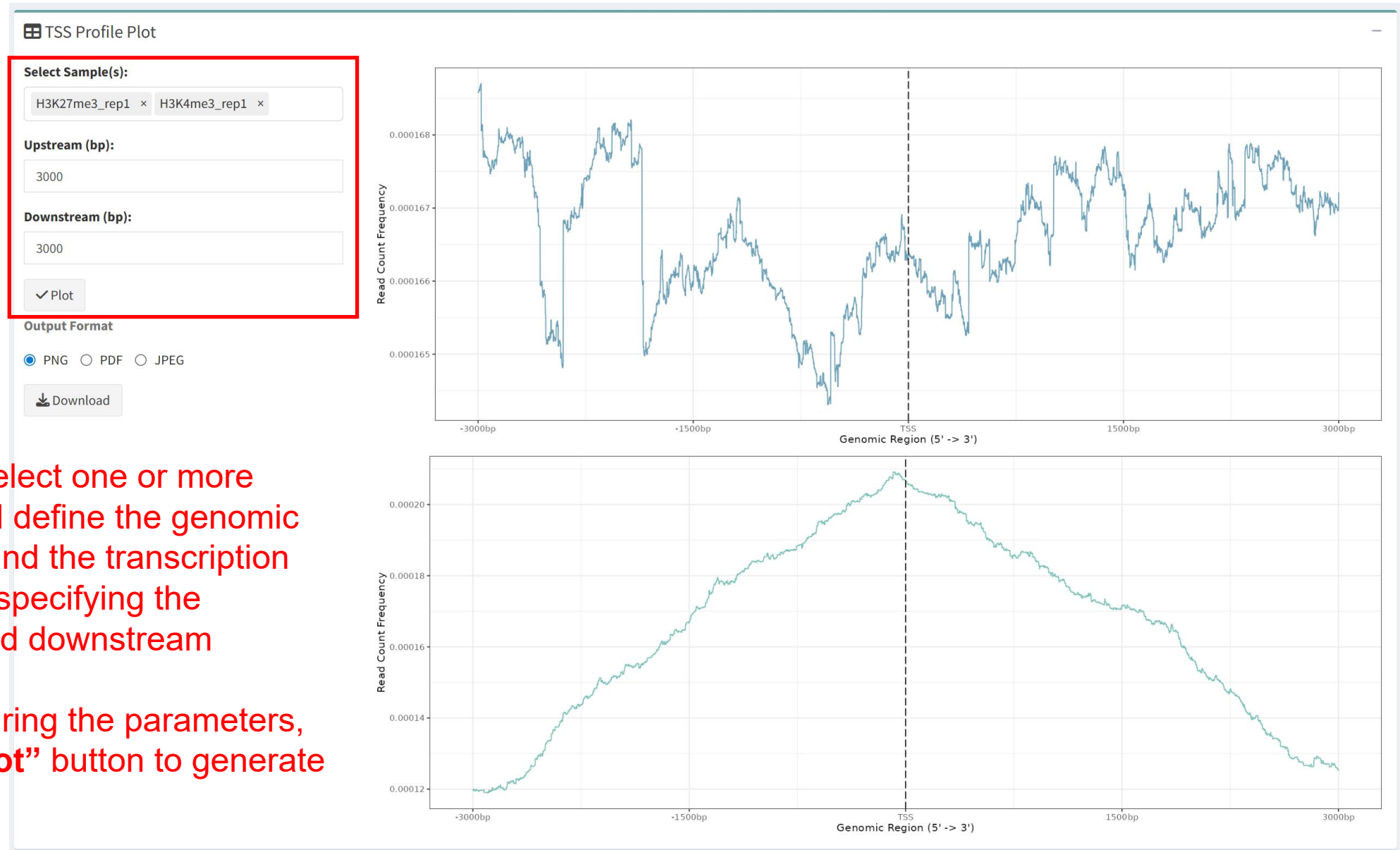
This step is optional and can be skipped.

Users may select one or more samples and specify a chromosome to visualize peak coverage. After making the selections, click the “**Plot**” button to generate the corresponding plot.



Peaks Visualization Module

This step is optional and can be skipped.



Users can select one or more samples and define the genomic window around the transcription start site by specifying the upstream and downstream distances. After configuring the parameters, click the “**Plot**” button to generate the plot.

Data Filtering and Grouping Module

Before Filter

After Filter

Count Threshold: ?

5

✓ Submit

Show 5 entries

Search:

hromosome	Start	End	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_rep1	H3K4me3_rep2
1r1	27805	30656	0	0	233	186
1r1	135727	140083	0	2	166	89
1r1	198335	200968	0	0	219	169
1r1	391014	392146	1	3	0	1
1r1	392617	393317	2	1	0	0

Showing 1 to 5 of 220,230 entries

Previous

1

2

3

4

5

...

44,046

Next

Users may specify a threshold value to filter the data.

During filtering, the system sums the read counts for each genomic region across samples and retains only those entries whose total count exceeds the defined threshold.

> 5

< 5

Data Filtering and Grouping Module

↔ Sample Grouping

Select Sample(s):

A549_4h_rep1 × A549_4h_rep2 × A549_4h_rep3 ×

Select the samples that should be grouped together.

Select Group Label:

Experimental group

Select “**experimental group**” or “**control group**” as the group name.

Add Group

Clear Groups

Use the “**Add Group**” button to create the group.

Use the “**Clear Groups**” button to remove all existing groups.

Current Groups:

\$Control
[1] "A549_0h_rep1" "A549_0h_rep2" "A549_0h_rep3"

All defined groups are listed under “**Current Groups.**”

✓ Submit

After all groups have been defined, click the “**Submit**” button to save the grouping information.

Differential Peak Analysis Module

⬅ Differential Analysis Parameters !

Direction of Change: ?

Both Directions ▼

Apply Independent Filtering for Low-Count Peaks

?
☒ Yes ☐ No

Independent filtering target adjusted p-value:

?
0.01

P-value adjustment method:

Benjamini–Hochberg (BH) ▼

▶ Run Differential Analysis

Direction of Change

This setting lets users choose whether to detect peaks that are increased or decreased between groups, allowing the analysis to focus on a specific direction of change.

Independent Filtering

Independent filtering is a DESeq2 step that ignores low-signal peaks during multiple-testing correction to improve detection power. Filtered peaks remain in the results, but their adjusted p-values may be NA.

target adjusted p-value

DESeq2 uses this value to choose a mean-count filtering threshold that helps maximize the number of significant peaks after multiple-testing correction. It does not directly define the final differential peaks; final peak selection is still based on the adjusted p-value and log2FC cutoffs specified later.

p-value Adjustment Method

This option allows users to select a method for multiple-testing correction.

- The Benjamini–Hochberg (BH) method controls the false discovery rate (FDR) and is well suited for analyses with many tests.
- The Holm method controls the family-wise error rate (FWER) and is more conservative, reducing false positives but potentially missing true signals.

Differential Peak Annotation Module

Input Annotation Parameters

Upstream (bp):

3000

Downstream (bp):

3000

▶ Run Peak Annotation

To annotate the peaks, first specify the upstream and downstream distances to define the genomic region of interest, and then click the “Run Peak Annotation” button to execute the annotation.

Differential Peak Visualization Module

 Volcano Plot

 MA Plot

 PCA Plot

 Heatmap

-log10(padj) Cutoff

1.3

Log2FC Threshold (Positive)

1

Log2FC Threshold (Negative)

1

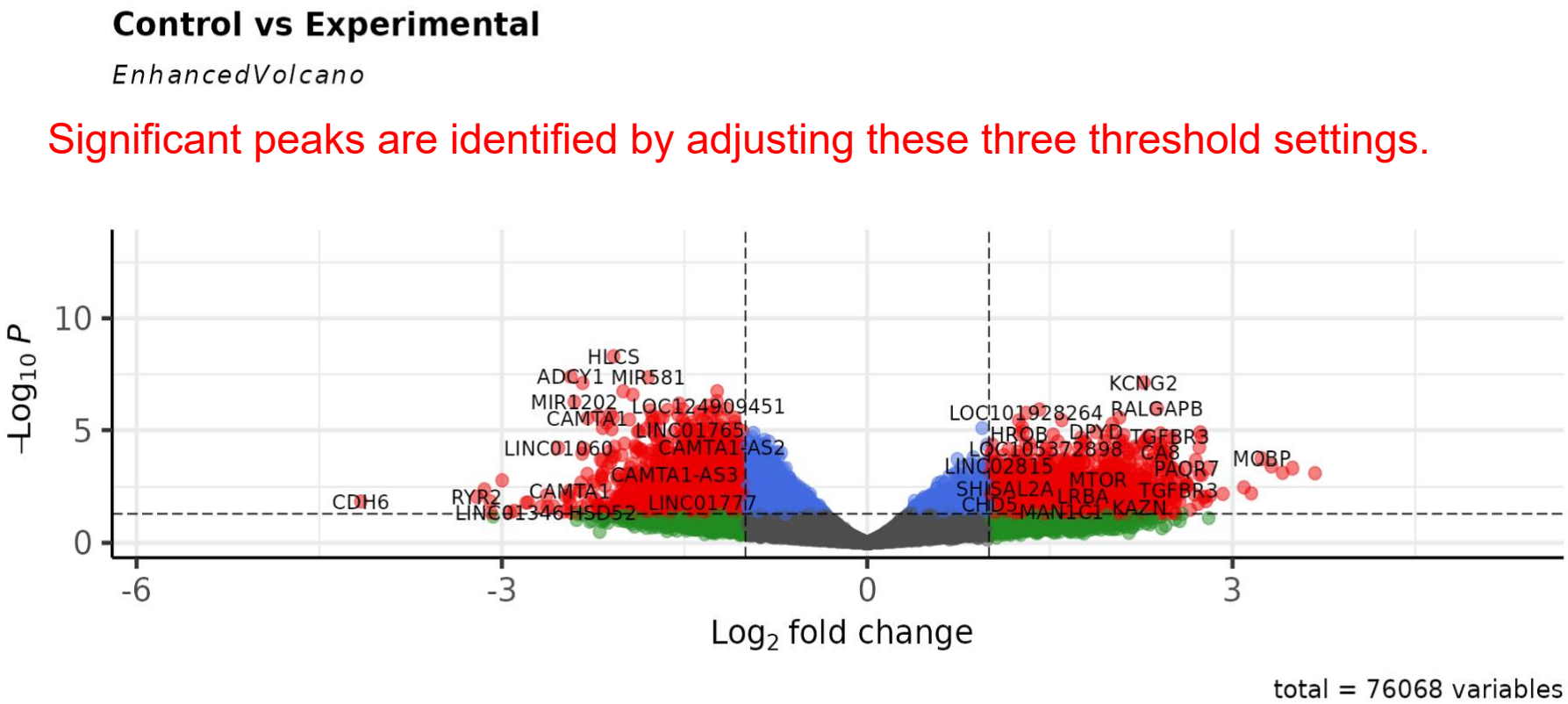
Point Size

3

Label Size

4

✓ Submit



Click the “Submit” button to update the plot and apply the selected significance criteria.

Differential Peak Visualization Module

Up Genes Preview

Upregulated Genes List

DVL1

MRPL20

NOL9

DNAJC11

DHRS3

RPL11

ELOA-AS1

SRRM1

RSRP1

MACO1

Show 5 entries

Search:

	seqnames	start	end	width	strand	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_
17	chr1	1346468	1350600	4133	*	0	0	
20	chr1	1405327	1408434	3108	*	0	0	
77	chr1	6552669	6555289	2621	*	0	0	
82	chr1	6699924	6702352	2429	*	0	0	
142	chr1	12614231	12620067	5837	*	0	0	

Showing 1 to 5 of 497 entries

Previous 1 2 3 4 5 ... 100 Next

Download

This section displays the upregulated and downregulated gene lists, as well as the peak tables generated from the filtering step.

Functional Enrichment Module

Select Enrichment Analysis Parameters ?

Analysis Type:

GO ▼

GO Ontology:

All ▼

p-value Cutoff:

0.05

q-value Cutoff:

0.05

Upregulated Genes List

CHD5
PARK7
LOC124903834
LINC01714
RERE
LOC124900441
LINC02606
SLC25A33
CASZ1
MTOR

Downregulated Genes List

LINC01346
LINC01777
CAMTA1-AS3
CAMTA1
CAMTA1-AS2
CAMTA1-AS1
MIR34A
VPS13D
DHRS3
TMEM51-AS2

Mixed-direction Genes List ?

MIR9-1HG
PBX1
KCNH1
CYP26B1
NPAS2
MIR4435-2HG
CDCA7
LOC102724861
MIR5702
COL4A3

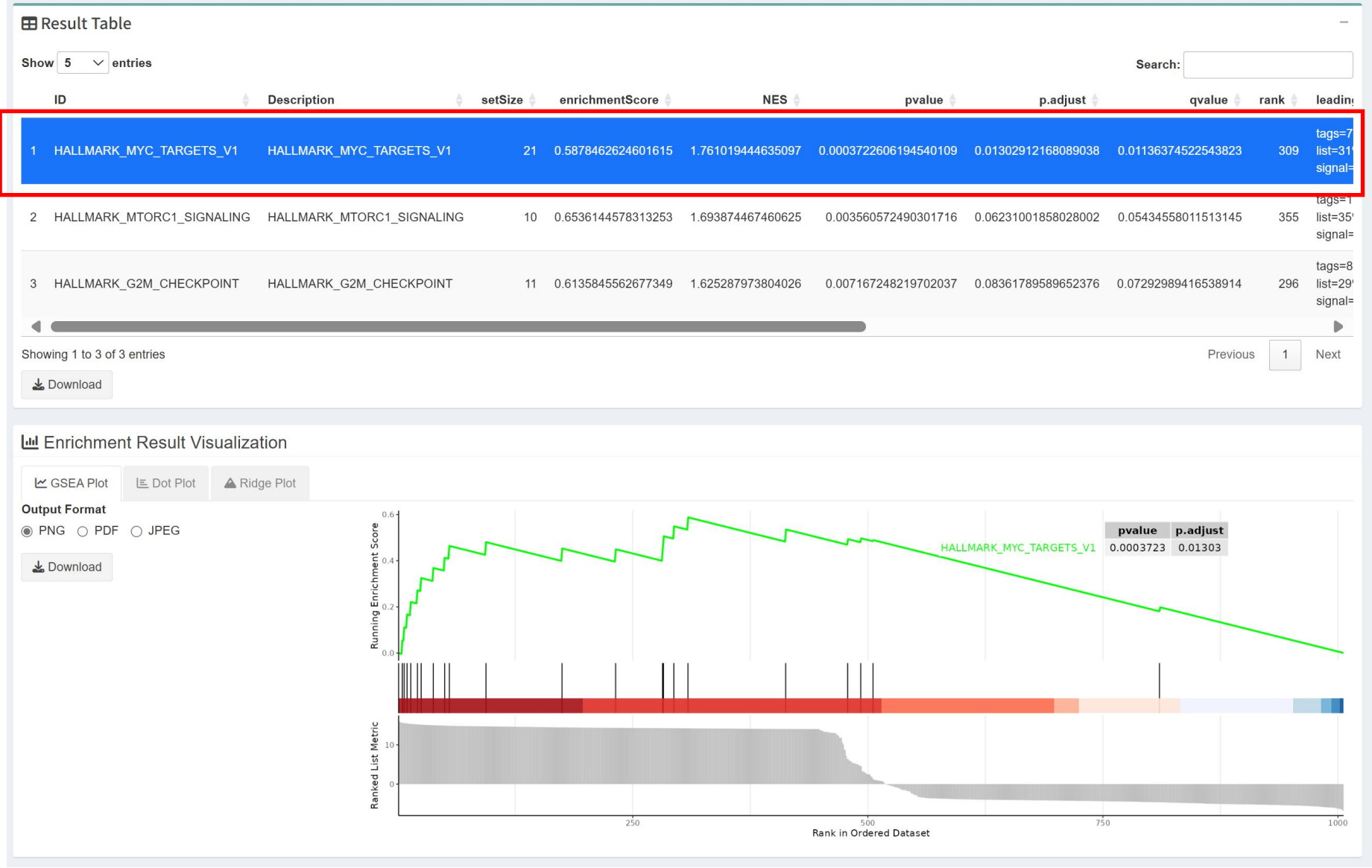
▶ Run Enrichment Analysis

Adjust the analysis type and corresponding parameters as needed.

For GO or KEGG analysis, users may enter or edit gene lists using gene symbols to define the input for enrichment testing. When GSEA is selected, the input is limited to the full set of genes annotated in the Gene Annotation module.

Genes in this list are linked to both increased-signal and decreased-signal differential peaks. Because their peak-level signals show conflicting directions, they are excluded from the direction-specific gene lists used for GO/KEGG ORA and from the default ranked gene list used for preranked GSEA. This list is provided for user inspection and transparency.

Functional Enrichment Module



When GSEA is selected, users must choose a specific enrichment entry before generating the corresponding GSEA plot.

Motif Enrichment Module

Upregulated Peaks List

chr3-150407290-150415877	▲
chr16-2150741-2157314	●
chr16-88972494-88978084	
chr6-119343897-119351533	
chr12-56682168-56690455	
chr8-38173859-38178882	
chr9-128688198-128692391	
chr11-65420641-65428391	
chr10-92688412-92694886	
chr6-27889944-27897392	▼
chr11-22222222-22222222	✎

Downregulated Peaks List

chr7-154746354-154754198	▲
chr11-20155235-20165969	●
chr7-27100472-27112603	
chr1-208072367-208087339	
chr1-8323134-8331215	
chr11-118150179-118161808	
chr20-47665285-47675999	
chr14-64734713-64750256	
chr1-28810214-28820062	
chr4-182814697-182830407	▼
chr12-118887818-11889817	✎

This section displays the top 200 upregulated and 200 downregulated peaks, which are selected from the differential peaks identified through the volcano plot filtering step. You can also input and modify your own lists of upregulated and downregulated peaks for further analysis.

Number of Top Transcription Factors to Plot

10

Display Motif GC Content

- ☒ Yes
- ☐ No

Enable Clustering

- ☒ Yes
- ☐ No

- Number of Top Transcription Factors to Plot:** Set how many top transcription factors you want to visualize (default is 10).
- Display Motif GC Content:** Choose whether to show the GC content of motifs.
- Enable Clustering:** Decide whether to enable clustering for grouping similar motifs in the plot.